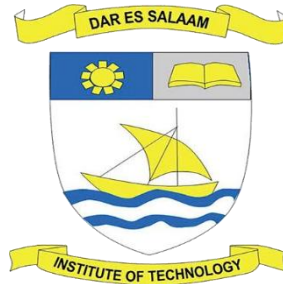


DAR ES SALAAM INSTITUTE OF TECHNOLOGY



DEPARTMENT OF COMPUTER STUDIES NTA LEVEL 8

MODULE NAME: PROJECT RELIZATION

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**PROJECT TITLE: BLOCKCHAIN-BASED NGO DONATION
MANAGEMENT SYSTEM**

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CHAPTER ONE: INTRODUCTION

1.1 Background Information

Non-Governmental Organizations (NGOs) play a vital role in supporting social and economic development through services such as education, healthcare, disaster relief, and poverty reduction. These organizations depend heavily on donations from individuals and institutions to fund their operations. With the rise of digital systems, many NGOs have moved toward online platforms to collect and manage donations more efficiently.

However, despite the adoption of digital solutions, issues related to transparency, trust, and accountability still exist. Donors often have limited visibility on how their contributions are used after donation. This creates uncertainty and reduces confidence in donation systems. Blockchain technology offers a solution by enabling secure, transparent, and tamper-proof recording of transactions, making it suitable for improving trust in NGO donation management systems.

1.2 Problem Statement

Most existing NGO donation systems lack full transparency in tracking how funds are managed and distributed. Donors typically receive confirmation of payment but cannot clearly see how their contributions are utilized or whether they reach the intended beneficiaries. This reduces trust and limits donor engagement over time.

Additionally, many systems rely on centralized databases that are vulnerable to manipulation, errors, and lack of real-time accountability. There is also limited integration with local payment methods such as mobile money, which are widely used in regions like Tanzania. Therefore, there is a need for a secure, transparent, and easily accessible donation management system that combines blockchain verification with mobile money integration to improve trust and accountability in NGO operations.

1.3 Main Objective

To design and develop a blockchain-based NGO Donation Management System that improves transparency, accountability, and secure management of donations through blockchain technology and local mobile money integration.

1.4 Specific Objectives

The project aims to:

1. To design a secure system for managing NGO fundraising campaigns and donor information.
2. To develop a transparent donation process that records donation and fund disbursement transactions on the blockchain.
3. To integrate local mobile money payment services to allow donors to contribute using familiar payment methods.
4. To provide real-time monitoring of donation progress, fund utilization, and beneficiary disbursement through interactive dashboards and notifications.
5. To develop an audit and reporting module that generates transaction logs, compliance reports.

1.5 Significance of the Project

The proposed system benefits several stakeholders.

NGOs

The system improves accountability by securely managing donations, beneficiaries, and campaign information while reducing manual record keeping.

Donors

Donors gain confidence through transparent tracking of donations, fund utilization, blockchain transaction verification, and real-time campaign updates.

Beneficiaries

The system ensures that approved funds are properly managed and distributed through monitored disbursement processes.

Researchers and Students

The project demonstrates the practical application of blockchain technology, web development, payment integration, and information systems in solving real-world problems.

Government and Regulatory Bodies

The system promotes transparency, financial accountability, and proper reporting, which support better governance of charitable organizations.

1.6 Scope of the Project

This project focuses on designing and developing a web-based Blockchain-Based NGO Donation Management System for registered NGOs. The system allows NGO administrators to create fundraising campaigns, manage beneficiaries, approve fund disbursements, and generate reports. Donors can register, browse campaigns, donate through local mobile money services, monitor campaign progress, view beneficiary disbursement summaries, receive notifications, and earn reward tokens based on their contributions.

The system records important donation and disbursement information on the blockchain to improve transparency and provide immutable transaction records. It also includes audit trails, reporting features, and a framework for future AI-based fraud detection.

The project does not implement cryptocurrency payments, cryptocurrency trading, or a complete machine learning fraud detection model. Instead, it focuses on blockchain transaction verification, secure donation management, and integration with local payment methods suitable for users in Tanzania.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

A literature review examines previous studies and existing technologies related to a research project. It helps identify the strengths and weaknesses of current systems and provides a basis for developing an improved solution. This chapter reviews existing donation management systems, identifies the research gap, and presents the proposed Blockchain-Based NGO Donation Management System.

2.2 Existing System

Many NGOs currently use traditional online donation platforms to collect and manage donations. These systems allow donors to contribute money through websites or online payment methods while administrators manually manage campaigns and financial records.

Although these systems simplify fundraising, they still have several limitations. Donation records are stored in centralized databases, making them vulnerable to unauthorized modification if proper security measures are not implemented. Donors also have limited visibility into how donated funds are distributed after reaching the organization. Most existing systems provide only basic donation confirmations without allowing donors to monitor beneficiary disbursements or view complete transaction histories.

Another limitation is that many blockchain-based donation platforms require cryptocurrency payments, making them difficult for ordinary users who are more familiar with local mobile money payment services.

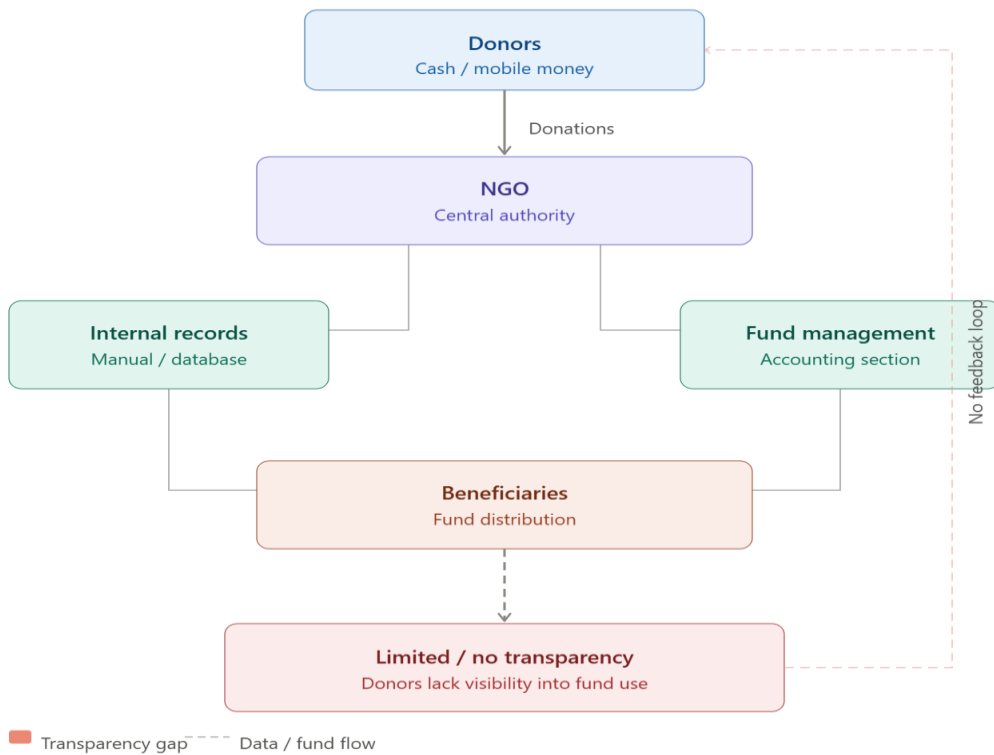


Figure 0-1 Existing Donation System

Limitations of Existing Systems

- Limited transparency after donations are made.
- Centralized storage of donation records.
- Limited donor access to fund utilization information.
- Weak audit capabilities.
- Limited support for blockchain verification.
- Many systems depend on cryptocurrency payments.
- Limited fraud monitoring mechanisms.

2.3 Research Gap

The review of existing systems shows that although online donation platforms improve fundraising, they do not fully address transparency, accountability, and trust between NGOs and donors. Most systems only allow donors to make contributions and receive payment confirmations without providing continuous updates regarding fund utilization and beneficiary support.

Furthermore, many existing blockchain donation platforms focus mainly on cryptocurrency transactions, making them less suitable for countries where mobile money services are the primary payment method. Existing systems also provide limited automated audit trails, compliance monitoring, and intelligent fraud detection.

These limitations create a need for a donation management platform that combines blockchain transparency with local payment methods while improving accountability through audit trails, reporting, and real-time notifications.

2.4 Proposed System

The proposed Blockchain-Based NGO Donation Management System addresses the identified challenges by integrating blockchain technology with a web-based donation platform. Instead of relying on cryptocurrency payments, donors can contribute using local mobile money services. Every donation and approved fund disbursement is securely recorded on the blockchain to provide an immutable transaction history.

The system allows NGO administrators to manage campaigns, beneficiaries, approvals, reports, and audit trails through a centralized administration panel. Donors can monitor campaign progress, view beneficiary disbursement summaries, receive real-time notifications, and verify donation records using blockchain transaction references. The proposed design also includes a reward token system that recognizes donor participation and prepares the platform for future AI-based fraud detection.

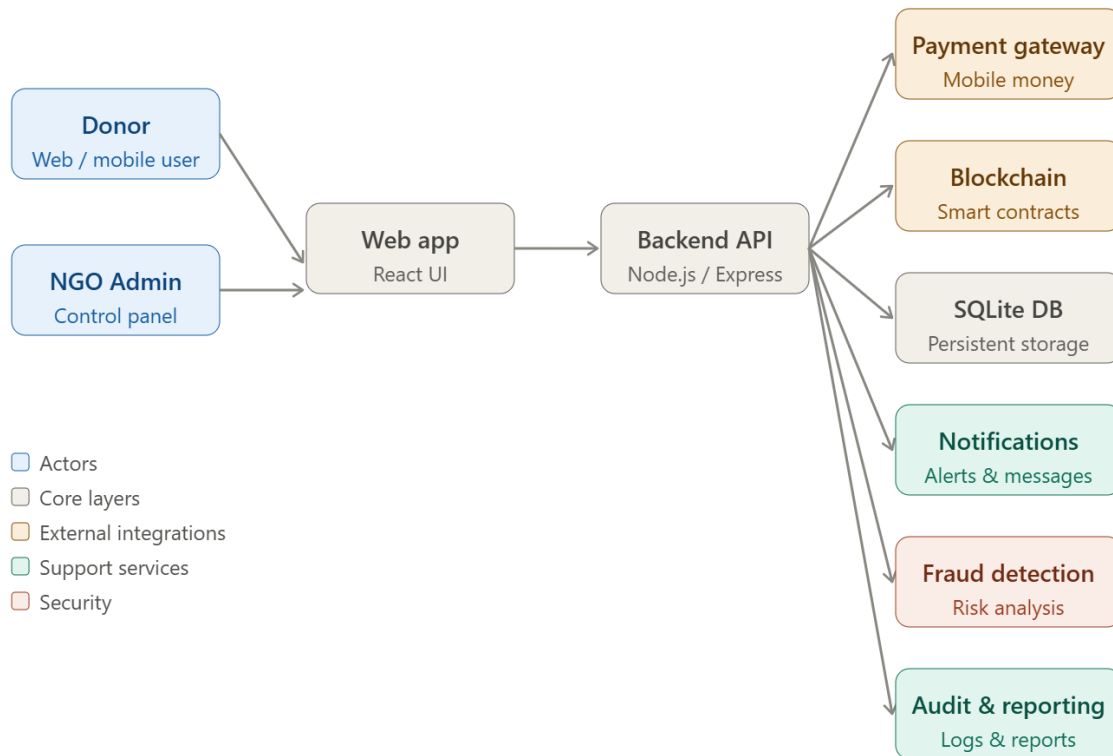


Figure 0-2 Proposed Donation System

Advantages of the Proposed System

- Blockchain-secured donation records.
- Mobile money payment integration.
- Transparent fund utilization.
- Beneficiary disbursement tracking.
- Automated audit trails.

- Compliance monitoring.
- Real-time notifications.
- Reward tokenization.
- Foundation for AI fraud detection.
- Improved donor confidence and accountability.

2.5 Chapter Summary

This chapter reviewed existing donation management systems and identified several challenges, including limited transparency, centralized data storage, poor audit capabilities, and limited support for local payment methods. The proposed Blockchain-Based NGO Donation Management System addresses these shortcomings by integrating blockchain technology, mobile money payments, audit trails, reporting, beneficiary tracking, and real-time notifications to improve transparency and accountability.

CHAPTER THREE: DATA ANALYSIS

3.1 Methodology

The project follows a **Agile development approach**, allowing iterative improvement of the system through continuous feedback and refinement. The system is designed using web technologies, blockchain integration, and mobile money payment services.

Tools used include:

- React (Frontend)
- Node.js / Express (Backend)
- SQLite (Database)
- Solidity / Smart Contracts (Blockchain)
- Web3 integration
- Google Forms (Questionnaire)

3.2 Data Analysis Overview

Data was collected from potential users including donors and NGO stakeholders to understand donation behaviour, transparency concerns, and system requirements. The analysis focuses on identifying patterns that support system design decisions.

A Google Form was shared with the Tanzania Red Cross Society cooperation chat groups, and their opinions were collected, the same was done with my fellow students to gather their opinions.

3.3 Data Gathering Methods

Questionnaires

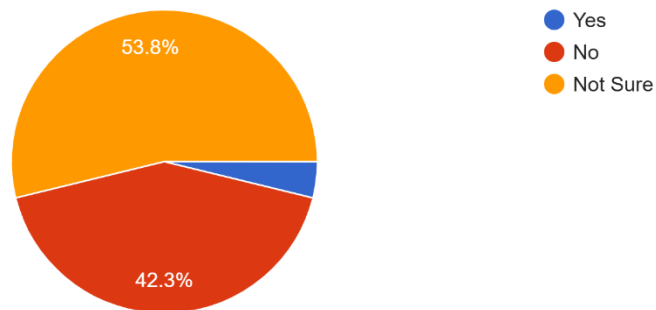
A structured questionnaire was used to collect user opinions on donation transparency, trust, and system usability.

Charts Analysis

Responses were analyzed using charts to identify trends in donor expectations and system needs.

Do you trust current NGO donation systems?

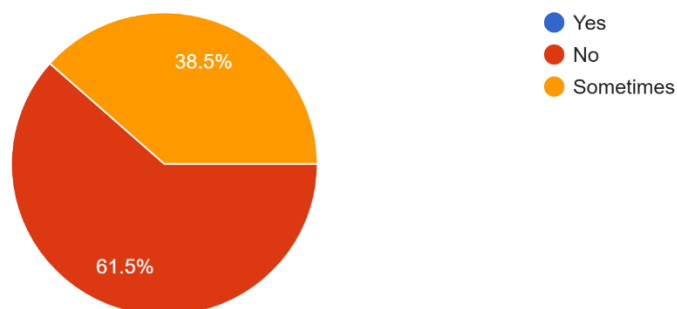
26 responses



Most respondents showed low or uncertain trust in current NGO donation systems. This suggests the need for a more transparent and reliable donation management system.

Can you easily track how your donation is used after donating?

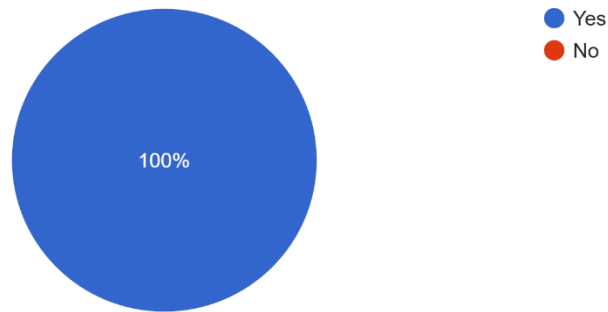
26 responses



The results indicate that many users cannot easily track their donations after donating. This highlights the need for a system that provides clear donation tracking and visibility.

Do you think NGOs should provide transparent donation tracking?

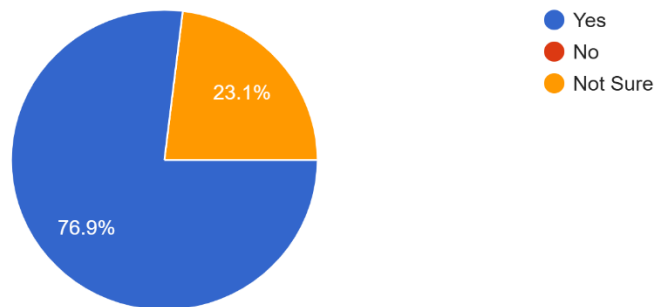
26 responses



Most respondents agreed that NGOs should provide transparent donation tracking. This supports the project's objective of improving transparency in donation management.

Do you think blockchain can improve transparency in NGO donations?

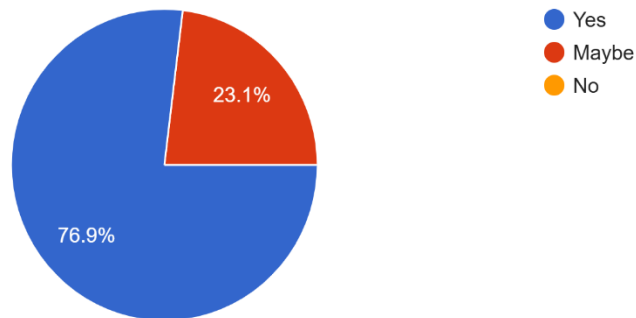
26 responses



The responses show positive support for using blockchain technology to improve transparency and secure donation records in NGO systems.

Do NGOs need better systems for managing donations?

26 responses



Most respondents believe NGOs need improved systems for managing donations. This finding supports the development of the proposed blockchain-based donation management system.

3.4 Questionnaire Summary

Based on responses:

- Most users prefer mobile money donations.
- Users want real-time tracking of donations.
- Transparency is a major concern for donors.
- Blockchain-based verification is considered useful for trust.

3.5 Summary of Findings

The analysis shows a strong need for a transparent donation system. Users prefer simple donation methods combined with real-time tracking and secure transaction verification. These findings support the development of a blockchain-based donation platform integrated with mobile money services.

CHAPTER FOUR: SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

4.1 Introduction

This section describes the software requirements of the Blockchain-Based NGO Donation Management System. It outlines the functional and non-functional requirements, system modules, and user interactions that define how the system operates.

4.2 Functional Requirements

The system shall:

- Allow users to register and log in securely.
- Enable donors to view and donate to NGO campaigns.
- Process donations through mobile money payment services.
- Record donation transactions on the blockchain for transparency.
- Allow NGO administrators to create and manage campaigns.
- Manage beneficiaries linked to specific campaigns.
- Process and approve fund disbursements.
- Provide real-time notifications for donation and disbursement events.
- Generate reports on donations, fund usage, and campaign progress.
- Maintain audit logs of all system transactions.

4.3 Non-Functional Requirements

- **Security:** The system shall protect user data and transactions using authentication and encryption.
- **Transparency:** All donations and disbursements shall be traceable through blockchain records.

- **Performance:** The system shall respond to user requests within a few seconds under normal conditions.
- **Reliability:** The system shall ensure accurate recording of all transactions without data loss.
- **Scalability:** The system shall support increasing users and campaigns without performance degradation.
- **Usability:** The interface shall be simple and easy to use for both donors and administrators.
- **Availability:** The system shall be accessible online at all times.
- **Maintainability:** The system shall be modular to allow future updates and improvements.

4.4 Module Functionality

The system consists of the following modules:

- **User Management Module:** Handles registration, login, and authentication.
- **Campaign Management Module:** Allows creation and monitoring of donation campaigns.
- **Donation Module:** Handles donor contributions and payment processing.
- **Beneficiary Module:** Manages individuals or groups receiving support.
- **Disbursement Module:** Controls release of funds to beneficiaries.
- **Blockchain Module:** Stores immutable records of transactions.
- **Notification Module:** Sends real-time updates to users.
- **Reporting Module:** Generates system reports and summaries.
- **Audit Module:** Tracks all transactions for accountability.

4.5 System Constraints

- The system requires internet connectivity to function.
- Mobile money integration depends on external payment APIs.

- Blockchain transactions may experience slight delays due to network processing.
- The system does not support cryptocurrency trading.
- AI-based fraud detection is not fully implemented, only designed for future enhancement.

4.6 Use Case Overview

The system involves two main actors: Donor and NGO Administrator.

- The donor views campaigns, makes donations, and tracks contributions.
- The administrator manages campaigns, beneficiaries, disbursements, and system reports.

A detailed use case diagram will be provided in the system design section.

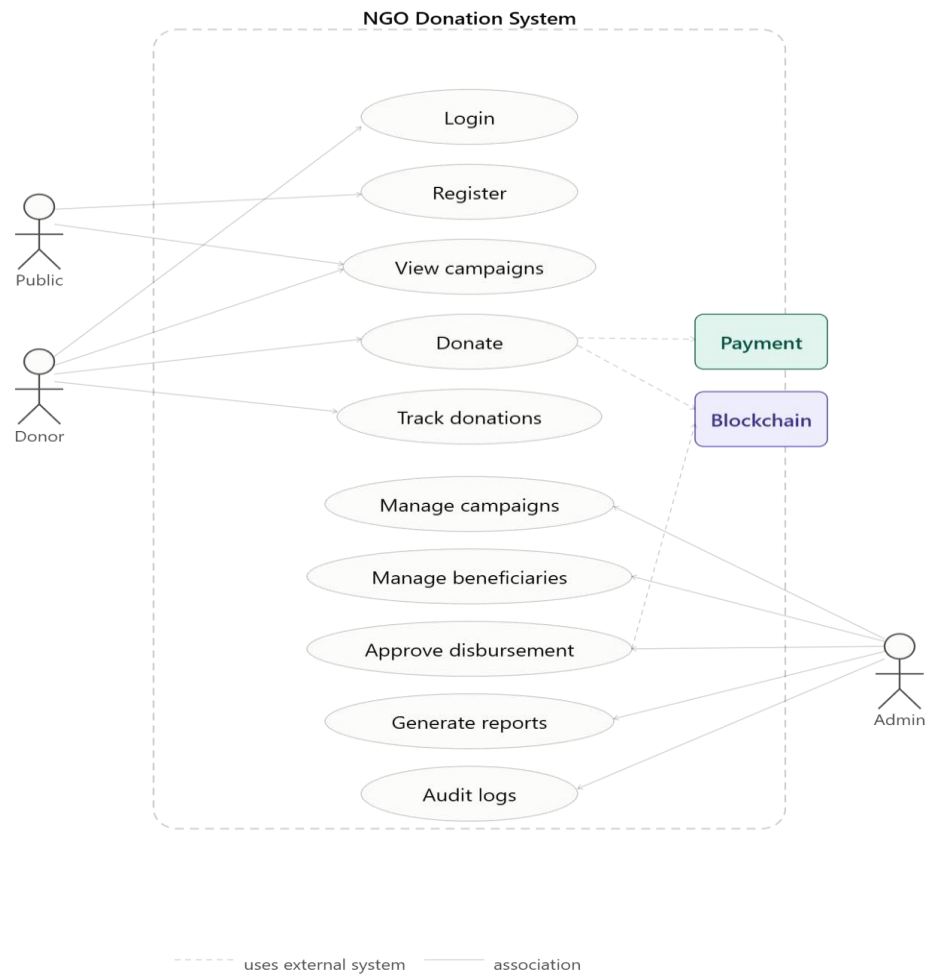


Figure 0-1 Use Case Diagram

CHAPTER 5: SYSTEM DESIGN

5.1. Introduction

The System Design document presents the proposed design of the Blockchain-Based NGO Donation Management System before implementation. It explains how the system will be structured, how information flows between different components, how data will be stored, and how users will interact with the platform. The design serves as a blueprint for system development and ensures that all functional requirements identified during the analysis phase are properly addressed.

The proposed system aims to improve transparency and accountability in NGO donation management by integrating blockchain technology with a web-based platform. The system enables donors to contribute funds through local mobile money services, allows NGO administrators to manage fundraising campaigns and beneficiaries, and records important transactions on the blockchain to provide immutable proof of donations and fund disbursements. The design also incorporates notification services, reporting tools, audit trails, and a future machine learning module for fraud detection.

5.2. Data Modelling

Data modelling describes how data enters the system, how it is processed, how it is stored, and how different entities relate to one another.

5.2.1 Data Flow Diagram (Level 0)

The Level 0 Data Flow Diagram, also known as the Context Diagram, provides a high-level view of the entire Blockchain-Based NGO Donation Management System. It illustrates the interaction between external entities and the system without exposing internal processes. The main external entities are the Donor, the NGO Administrator, the Mobile Money Payment Gateway, and the Blockchain Network. Donors submit donations and receive confirmations, while administrators manage campaigns, beneficiaries, and reports. The payment gateway processes financial

transactions, and the blockchain network stores immutable records of verified donations and disbursements.

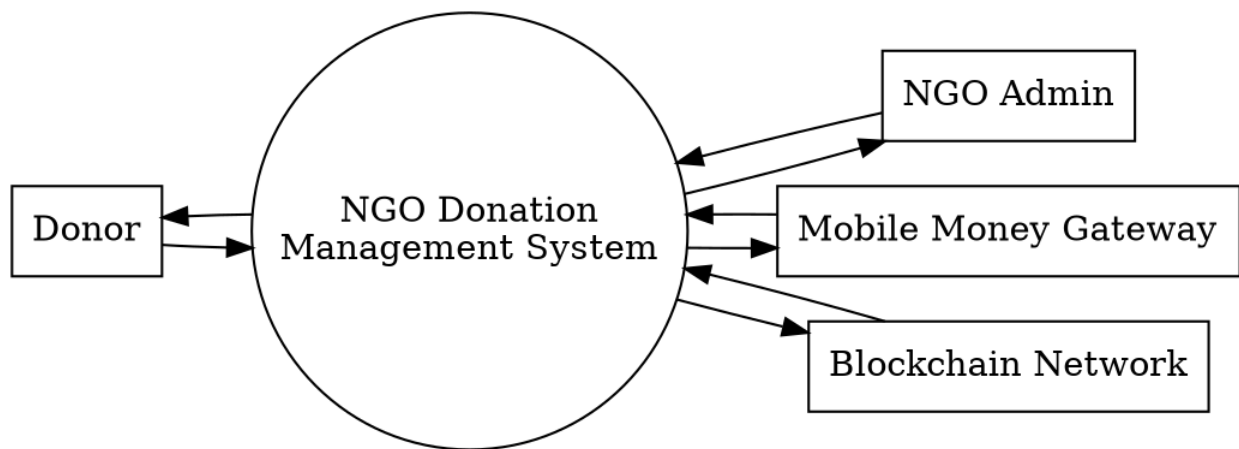


Figure 0-1 Data Flow Diagram Level 0

5.2.2 Data Flow Diagram (Level 1)

The Level 1 Data Flow Diagram expands the Context Diagram by dividing the system into its major functional processes. These processes include User Authentication, Campaign Management, Donation Processing, Beneficiary Management, Disbursement Management, Blockchain Recording, Audit and Reporting, and Notification Management. The diagram also illustrates how data is stored in the system database and how blockchain records are created after successful transactions.

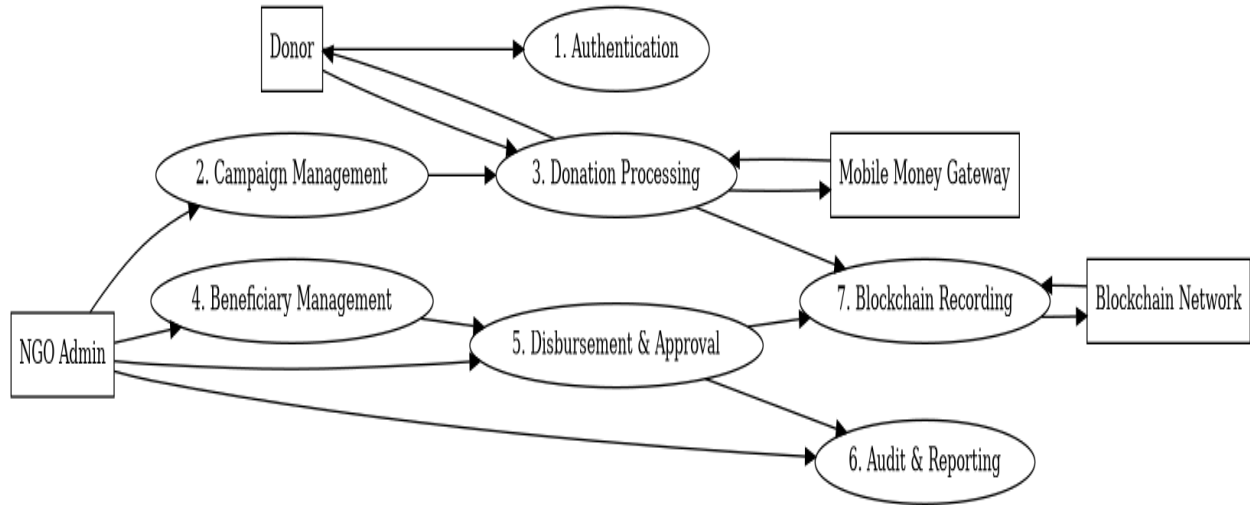


Figure 0-2 Data Flow Diagram Level 1

5.2.3 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram illustrates the structure of the system database by showing the entities, their attributes, and the relationships between them. The proposed database consists of Users, Campaigns, Donations, Beneficiaries, Disbursements, Approvals, Notifications, Reward Tokens, and Fraud Alerts. These entities work together to support campaign management, donation processing, beneficiary tracking, reporting, and audit functions.

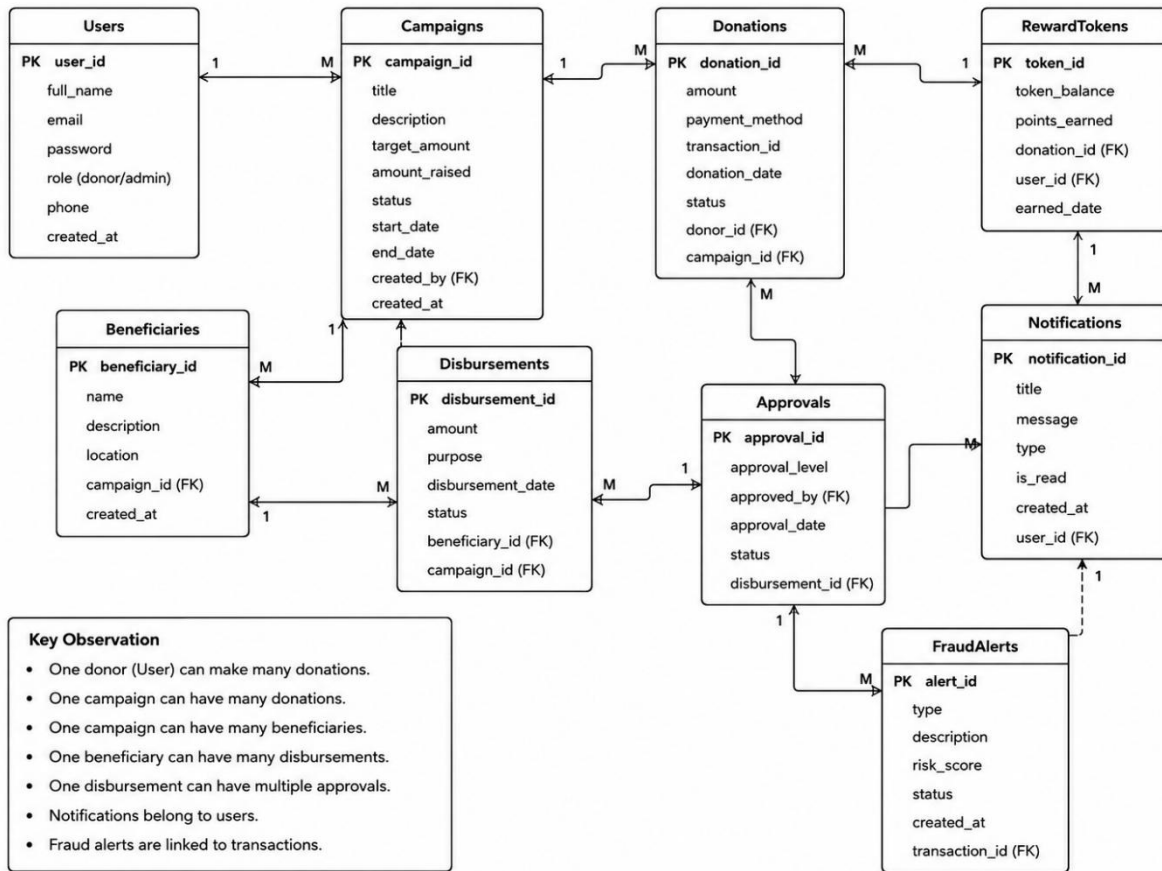


Figure 0-3 Entity Relationship Diagram

5.2.4 Blockchain Data Model

Unlike the relational database, blockchain is only used to store immutable transaction records that require transparency and integrity. Instead of storing complete user information, the blockchain stores transaction identifiers, timestamps, digital signatures, blockchain hashes, campaign identifiers, approval status, and transaction amounts. This approach improves security while ensuring that important financial records cannot be altered after they have been verified.

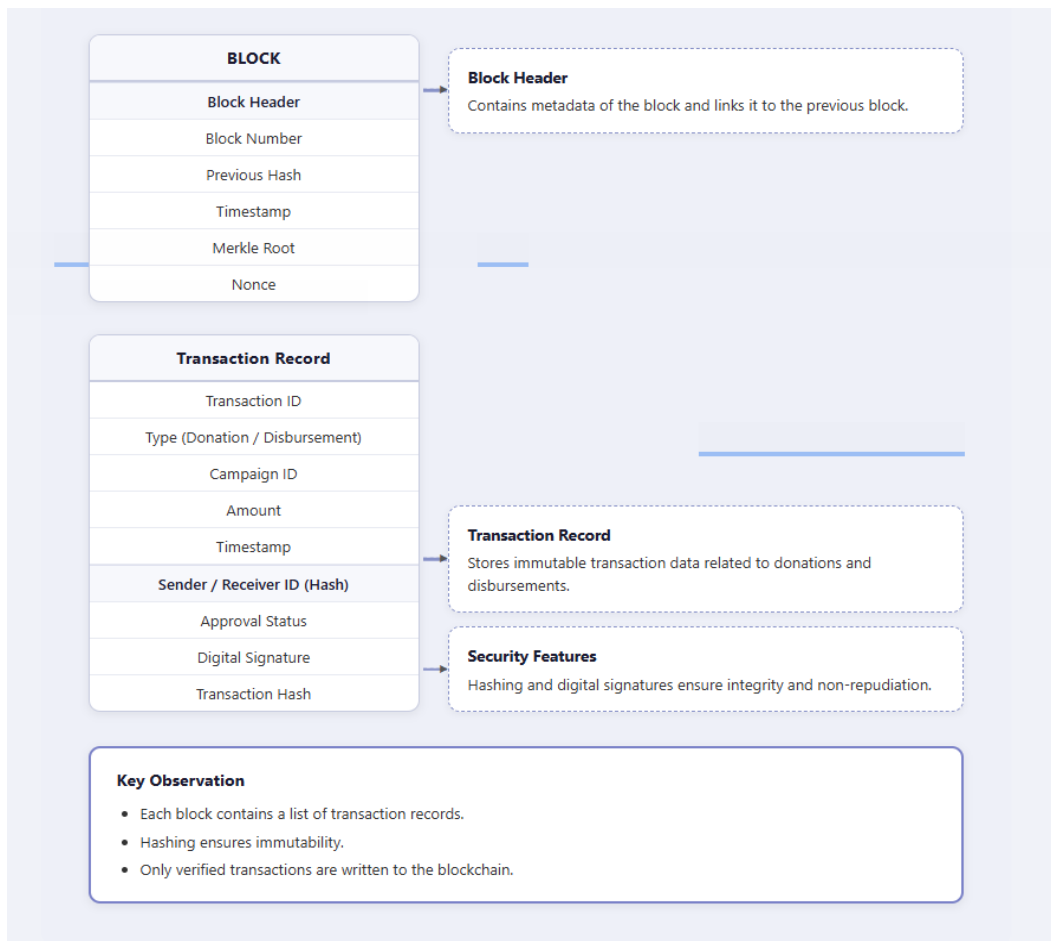


Figure 0-4 Blockchain Data Model

5.2.5 Machine Learning Dataset

The Machine Learning Dataset is designed to support future implementation of AI-based fraud detection. Although the current prototype uses rule-based fraud detection, the collected transaction data will later be used to train machine learning models capable of identifying suspicious donation patterns. The dataset includes transaction amount, donor identifier, campaign identifier, payment method, transaction time, blockchain verification status, and fraud labels.

Dataset: Transaction_Fraud_Dataset

Field Name	Description	Type	Example
transaction_id	Unique transaction identifier	String	TXN12345
donor_id	Hashed donor identifier	String	DNR_8f3d2a
campaign_id	Campaign identifier	String	CMP_0012
amount	Donation amount	Float	50.00
payment_method	Payment method used	String	Mobile Money
transaction_time	Date and time of transaction	DateTime	2024-05-20 14:32:10
location	Donor location (if available)	String	Kampala
device_type	Device used	String	Android
ip_address	IP address of transaction	String	102.168.1.5
is_blockchain_verified	Blockchain verification status	Boolean	True
fraud_label	Target label (Fraud / Not Fraud)	String	Not Fraud
risk_score	Fraud risk score (0 – 1)	Float	0.12

Key Observation

- Dataset contains historical transaction data.
- Used to train ML models for fraud detection.
- *fraud_label* is the target variable.

Figure 0-5 Machine Learning Dataset

5.3. Process Modelling

Process modelling illustrates how users perform activities within the system from beginning to end.

5.3.1 Donor Activity Diagram

The Donor Activity Diagram illustrates the complete donation process. The donor logs into the system, browses available campaigns, selects a campaign, enters the donation amount, completes

payment using a supported mobile money service, and receives a transaction confirmation. Once payment is verified, the backend records the donation in the database, stores proof on the blockchain, updates campaign progress, awards reward tokens, and notifies the donor.

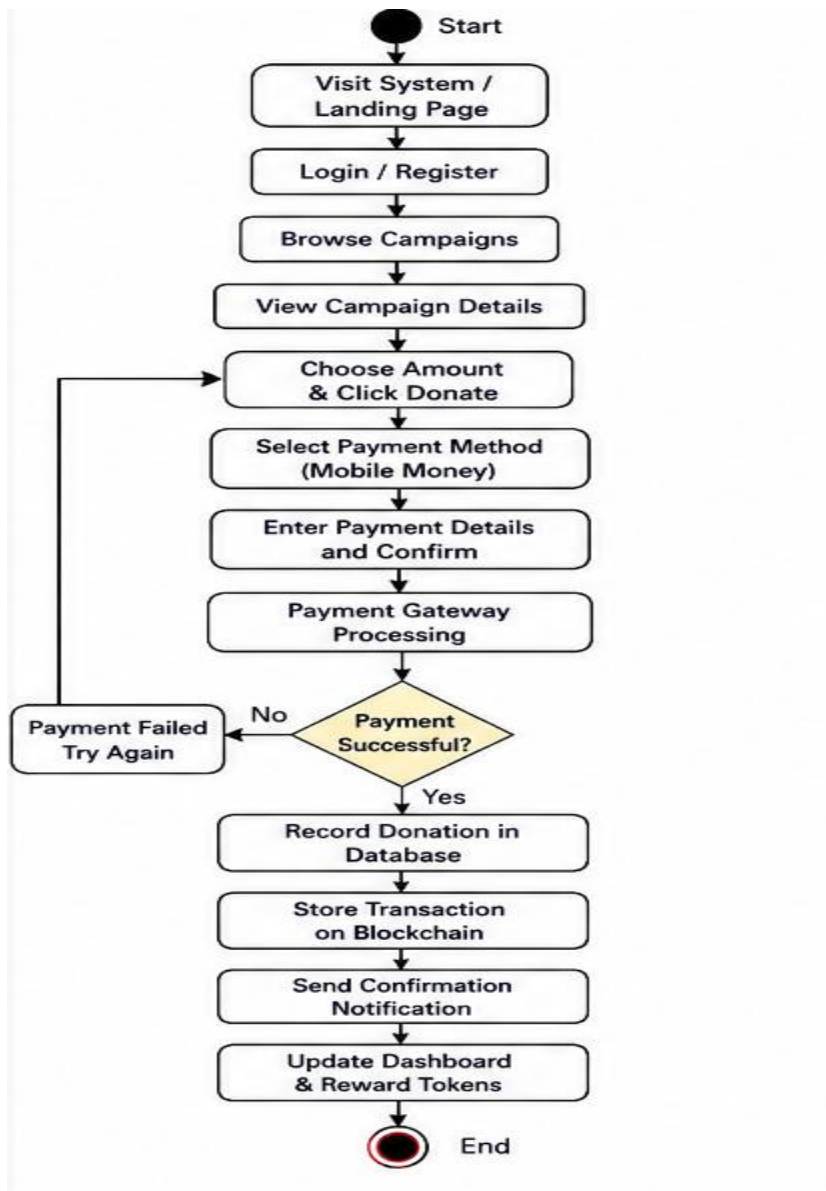


Figure 0-6 Donor Activity Diagram

Key Observation

This activity demonstrates how transparency is achieved by combining payment verification, blockchain recording, and real-time notifications.

5.3.2 Administrator Activity Diagram

The Administrator Activity Diagram illustrates how NGO administrators manage fundraising campaigns and beneficiaries. After authentication, administrators create campaigns, register beneficiaries, monitor donations, approve fund disbursement requests, generate reports, and monitor fraud alerts. The administrator also reviews audit logs to ensure transparency and accountability throughout the donation process.

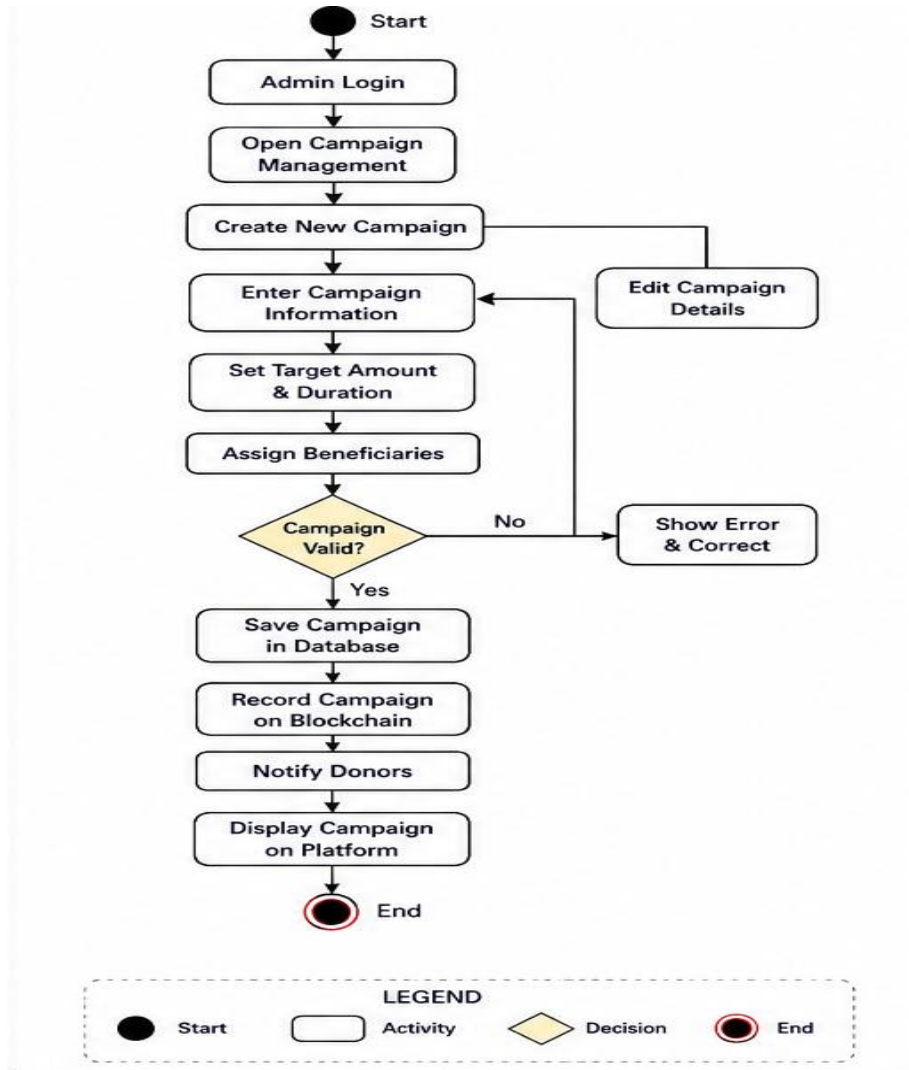


Figure 0-7 Administrator Activity Diagram

Key Observation

Administrative activities focus on campaign management, monitoring, approval, and reporting.

5.3.3 Disbursement Approval Activity Diagram

The Disbursement Approval Activity Diagram illustrates how funds are released to beneficiaries. The administrator initiates a disbursement request, which undergoes approval checks. Large transactions require multiple approvals before the smart contract authorizes the release of funds. Once approved, the blockchain records the transaction, notifications are generated, and reports are automatically updated.

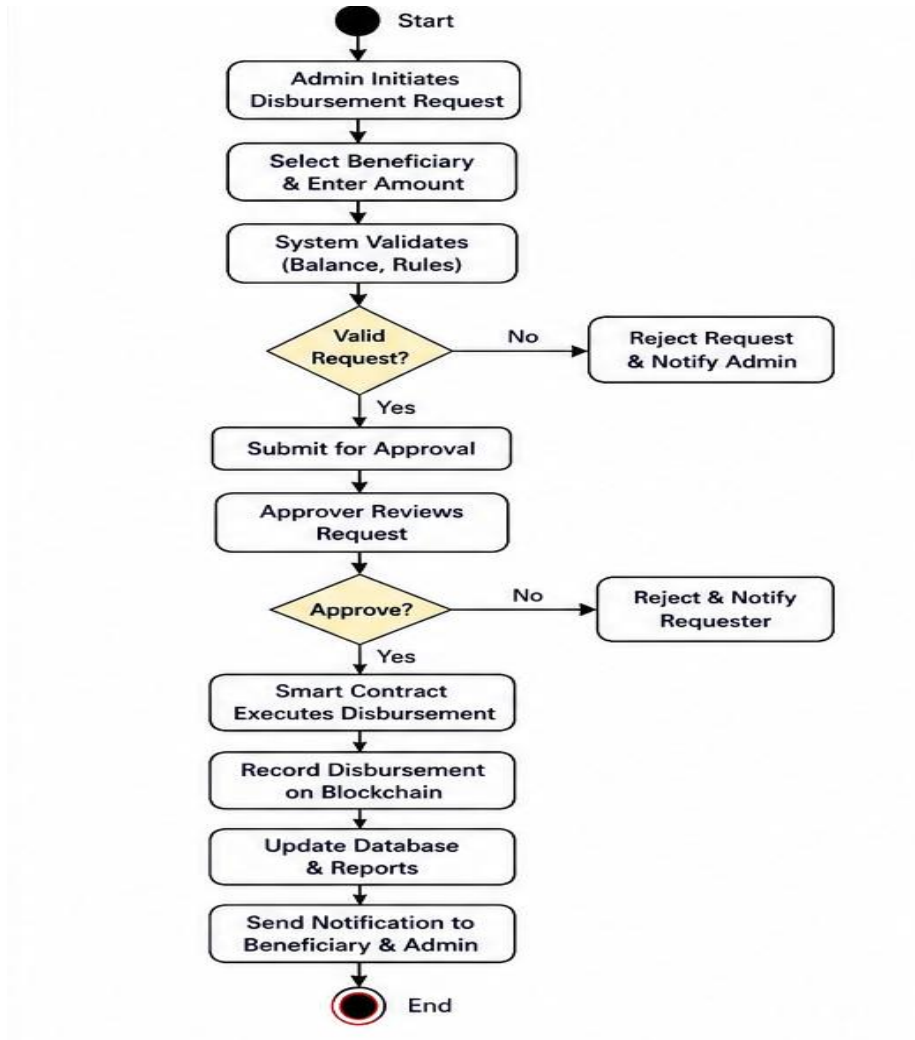


Figure 0-8 Disbursement Approval Activity Diagram

Key Observation

The approval workflow ensures governance, accountability, and secure fund utilization.

5.4. User Interface Design

The user interface has been designed to provide a simple, user-friendly, and transparent experience for both donors and administrators. The design emphasizes ease of navigation while ensuring that users can quickly access the features relevant to their responsibilities.

5.4.1 Landing Page

The Landing Page serves as the public entry point of the system. It introduces the platform, displays featured fundraising campaigns, and provides access to login and registration pages. Visitors can view campaign summaries before creating an account or signing in.

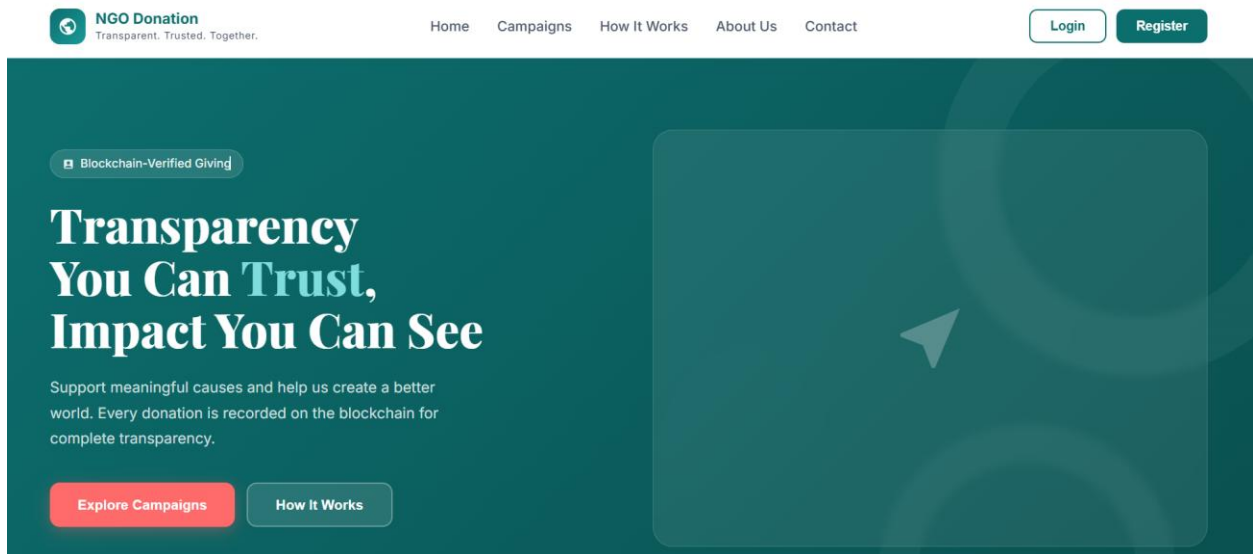


Figure 0-9 Landing Page

5.4.2 Campaign Details Page

The Campaign Details Page provides complete information about a selected fundraising campaign. Users can view the campaign description, fundraising target, donation progress, beneficiary information, recent fund disbursements, and the donation section. This page serves as the primary interface where donors decide whether to contribute.

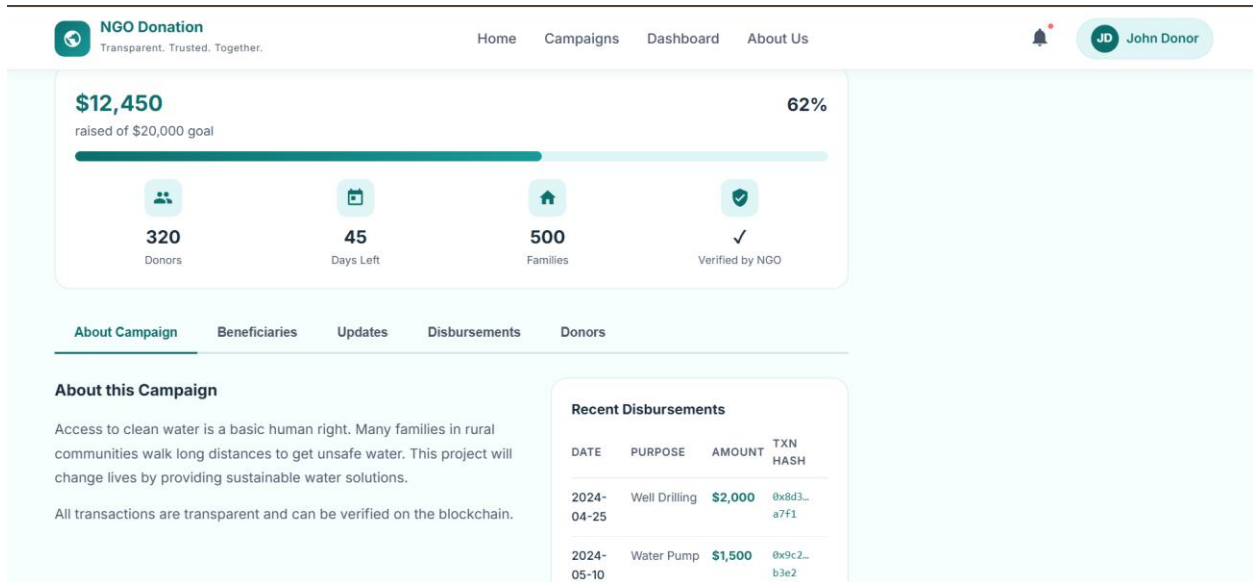


Figure 0-10 Campaign Details Page

5.4.3 Donation Page

The Donation Page enables donors to contribute funds using supported mobile money payment services. After entering the donation amount and payment details, the system verifies the transaction, records the donation, generates a blockchain proof, updates campaign progress, awards reward tokens, and displays a confirmation receipt

YOU ARE DONATING TO:

Clean Water for Village

Water & Sanitation

Campaign Progress: 62%

\$12,450 of \$20,000

Transaction will be recorded on the blockchain for full transparency.

Enter Donation Details

AMOUNT (USD)

\$ 50.00

\$10 \$25 \$50 \$100

PAYMENT METHOD

Credit Card

SELECT MOBILE MONEY PROVIDER

Airtel Money

MOBILE NUMBER

0783430151

Proceed to Pay

How it works?

STEP 1 You Donate
Choose amount and pay using mobile money.

STEP 2 We Verify
Payment is verified and recorded in our system.

STEP 3 Blockchain Record
Transaction is stored on the blockchain.

STEP 4 Impact Created
Funds are used to help communities.

Figure 0-11 Donation Page

5.4.4 Donor Dashboard

The Donor Dashboard provides a personalized overview of the donor's activities within the system. It displays donation history, campaigns supported, reward token balance, recent notifications, and quick access to donation records. Although donors are redirected to the Campaign List immediately after login, the dashboard remains available from the navigation menu as a personal summary page.

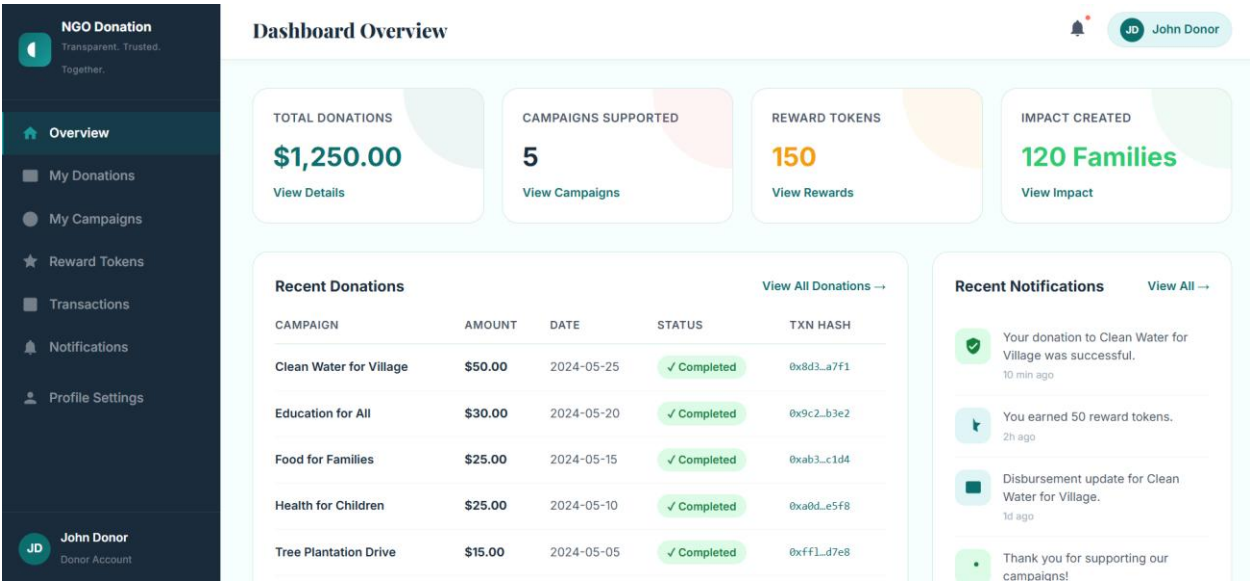


Figure 0-12 Donor Dashboard

5.4.5 Administrator Dashboard

The Administrator Dashboard functions as the operational control center of the platform. It provides real-time summaries of active campaigns, donations received, fund disbursements, pending approvals, fraud alerts, audit logs, and generated reports. Administrators use this

dashboard to monitor the overall performance and transparency of the NGO donation management system.

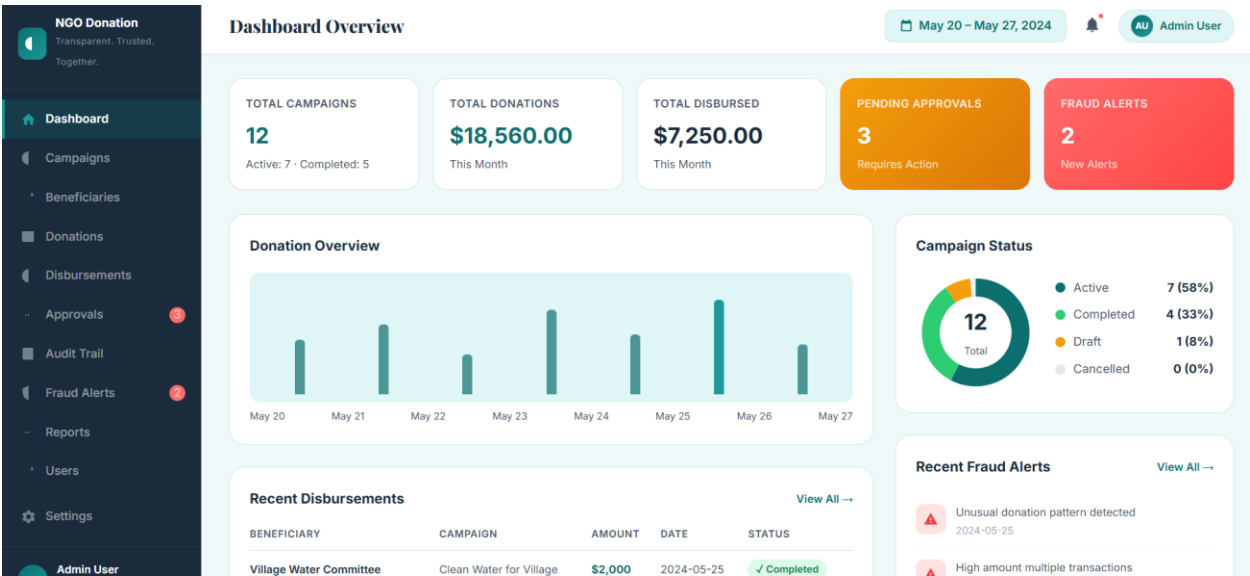


Figure 0-13 Administrator Dashboard

CHAPTER SIX: CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

This project presents a Blockchain-Based NGO Donation Management System designed to improve transparency, accountability, and efficiency in the management of charitable donations. The system addresses key challenges faced by traditional NGO donation platforms, such as lack of visibility into fund usage, limited trust between donors and organizations, and inefficient tracking of donations and disbursements.

By integrating blockchain technology, the system ensures that all donation and disbursement transactions are securely recorded and cannot be altered. This improves trust and provides an immutable audit trail of financial activities. The integration of mobile money payment systems makes the platform accessible to users in Tanzania without requiring cryptocurrency knowledge.

The system also introduces real-time notifications, audit logs, reporting tools, and a structured approval process for fund disbursement. Additionally, a foundation for AI-based fraud detection is included to enhance future system intelligence and security.

Overall, the proposed system provides a modern and transparent approach to managing NGO donations while improving donor confidence and operational accountability.

6.2 Recommendations

The following recommendations are made for future improvement of the system:

1. The system should be extended to include a fully developed AI-based fraud detection model to automatically detect suspicious transactions.
2. Mobile application versions (Android and iOS) should be developed to increase accessibility and usability for donors and administrators.
3. Integration with additional payment gateways beyond mobile money should be considered to support international donors.

4. The blockchain layer should be expanded to support more advanced smart contract automation for disbursement and governance processes.
5. Future versions of the system should include advanced analytics dashboards to provide deeper insights into donation trends and impact measurement.

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